

First-Year First-Semester Exam

January 2025

Answer the following problems and show all your work. Support all statements to the best of your ability. Use a separate sheet of paper for each problem. Each is worth 10 points.

1. Show that for every set X , there is no function $f : X \rightarrow 2^X$ such that f is onto. Note that 2^X is the power set on X and is also denoted by $\mathcal{P}(X)$.
2. Let $\mathbb{R}^\infty$ denote the subset of $\mathbb{R}^\omega$ consisting of all sequences that are “eventually zero,” that is, all sequences $(x_1, x_2, \dots)$ such that $x_i \neq 0$ for only finitely many values of i . What is the closure of $\mathbb{R}^\infty$ in the box and product topologies? Justify your answer.
3. Let $p : X \rightarrow Y$.
 - (a) Define what it means for p to be a quotient map.
 - (b) Show that if there is a continuous map $f : Y \rightarrow X$ such that $p \circ f$ equals the identity map of Y , then p is a quotient map.
4.
 - (a) Give the topology of the one-point compactification of a locally compact Hausdorff topological space X .
 - (b) Show that the one-point compactification of $\mathbb{Z}_+$ is homeomorphic with the subspace $\{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{Z}_+\}$ of $\mathbb{R}$.
5. Let (X, d) be a complete metric space. State and prove the contraction mapping theorem for X , which is also called the Banach fixed point theorem.

Answer the following with complete definitions or statements or short proofs. Each is worth 5 points.

6. Show that the axiom of choice is equivalent to the statement that for any indexed family $\{A_\alpha\}_{\alpha \in J}$ of nonempty sets with $J \neq \emptyset$, the cartesian product $\prod_{\alpha \in J} A_\alpha$ is nonempty.
7. Show that a connected normal space X having more than one point is uncountable.
8. Suppose that $f : X \rightarrow Y$ is continuous and that $A \subset X$ is connected. Show that $f(A) \subset Y$ is connected.
9. Suppose that $f : X \rightarrow Y$ is continuous and that $A \subset X$ is compact. Show that $f(A)$ is compact.
10. State the Extreme Value Theorem.
11. Define what it means for a topological space to be regular.
12. What does it mean for a topological space X to be a Baire space?
13. State the Cantor-Schröder-Bernstein theorem.
14. Is every path connected space locally path connected?
15. Define what it means for a topological space X to be separable. Let I denote the unit interval $[0, 1]$. Is $I^{\mathbb{Z}_+}$ separable? What about I^I ?